

Paper 1 - Set 12 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	E	Trigonometric functions and equations
2	B	Integration
3	D	Ratio, proportion and percentages
4	F	Circles and circle theorems
5	C	Functions and transformations
6	D	Number, units and estimation
7	B	Exponentials and logarithms
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9	B	Trigonometric functions and equations
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13	A	Simultaneous equations and inequalities
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17	A	Differentiation
18	F	Quadratic functions
19	A	Integration
20	C	Binomial expansion

Worked solutions

Question 1 TRG-11 · Trigonometric functions and equations · spec MM4.4

The function $f(x) = 3\sin(2x) - 1$ is defined for all real x . Find the difference between its maximum and minimum values.

Answer: E

Range is $[-4, 2]$; the difference is 6.

Full method - how and why

Range of $a\sin(bx) + c$. Only a and c matter - the b affects the period, not the values taken.

- 1 $\sin(2x)$ takes every value in $[-1, 1]$.

why: Compressing horizontally changes how fast the wave oscillates, not how high it goes.

- 2 Multiply by 3: the range becomes $[-3, 3]$.

why: Amplitude $|a| = 3$.

- 3 Subtract 1: the range becomes $[-4, 2]$.

why: The whole wave shifts down by 1.

- 4 Difference between maximum and minimum: $2 - (-4) = 6$.

why: Equivalently $2 \times \text{amplitude} = 2 \times 3 = 6$ - the vertical shift cancels out of a difference.

Where it goes wrong

- Answering 3 (the amplitude) instead of the full range $2a$.
- Letting the -1 affect the difference, giving 7 or 5.

Takeaway Max minus min for $a\sin(\cdot) + c$ is always $2|a|$; vertical shifts cancel.

Question 2 INT-06 · Integration · spec MM7.5

The trapezium rule with 2 strips of equal width is used to estimate $\int_0^2 x^2 dx$. Which of the following is true?

Answer: B

$h = 1$: $\frac{h}{2}[y_0 + 2y_1 + y_2] = \frac{1}{2}[0 + 2(1) + 4] = 3$. The true value is $\frac{8}{3}$; the curve bends upwards (convex), so the chords lie above it and the rule overestimates.

Full method - how and why

Apply the trapezium rule and then decide over- or under-estimate from the curve's shape.

- 1 Two strips on $[0, 2]$ means $h = 1$ and ordinates at $x = 0, 1, 2$: $y = 0, 1, 4$.

why: $h = \frac{b-a}{n}$ with n the number of strips (not the number of ordinates).

- 2 Apply the rule: $\frac{h}{2}[y_0 + 2y_1 + y_2] = \frac{1}{2}[0 + 2(1) + 4] = 3$.

why: Only the interior ordinates are doubled.

- 3 Compare with the exact value $\int_0^2 x^2 dx = \frac{8}{3} \approx 2.67$.

why: The estimate 3 is larger.

- 4 Explain it: $y = x^2$ bends upwards (convex), so each chord lies **above** the curve and the trapezia overestimate.

why: Convex $\Rightarrow$ overestimate; concave $\Rightarrow$ underestimate. This is the reasoning the question wants.

Where it goes wrong

- Forgetting to double the interior ordinate, giving $\frac{5}{2}$ - offered twice as a distractor.
- Saying 'underestimate' because the trapezia look like they cut corners - for a convex curve they do the opposite.

Takeaway Trapezium rule: $\frac{h}{2}[y_0 + 2(\text{middles}) + y_n]$; convex curves are overestimated.

Question 3 RAT-01 · Ratio, proportion and percentages · spec M3.8

The price of a coat is reduced by 20% in a sale, and then the sale price is reduced by a further 25%. The final price is what percentage of the original price?

Answer: D

$$0.8 \times 0.75 = 0.6, \text{ so } 60\%.$$

Full method - how and why

Successive percentage changes **multiply**. Convert each to a multiplier and combine.

- 1 A 20% reduction multiplies by 0.8.
why: You keep 80% of the price: $1 - 0.20 = 0.8$.
- 2 A further 25% reduction multiplies by 0.75 - applied to the **sale** price, not the original.
why: The second discount acts on whatever the price already is; that is why the changes multiply.
- 3 Combine: $0.8 \times 0.75 = 0.6$.
why: Multiplying the multipliers gives the overall multiplier.
- 4 So the final price is 60% of the original.
why: Equivalently a 40% reduction overall - not 45%.

Where it goes wrong

- Adding the percentages to get 45% off, hence 55% - an offered option.
- Applying the second reduction to the original price.

Takeaway Percentage changes compose by multiplying multipliers, never by adding percentages.

Question 4 CIR-08 · Circles and circle theorems · spec MM3.3

A chord of a circle of radius 13 is at perpendicular distance 5 from the centre. Find the length of the chord.

Answer: F

The perpendicular from the centre bisects the chord: half-length = $\sqrt{13^2 - 5^2} = 12$, so the chord is 24.

Full method - how and why

The perpendicular from the centre bisects the chord, creating a right-angled triangle with the radius as hypotenuse.

- 1 Draw the radius to one end of the chord and the perpendicular from the centre: they form a right-angled triangle.
why: The perpendicular distance, half the chord and the radius are the three sides.
- 2 Apply Pythagoras: half-chord = $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.
why: The radius 13 is the hypotenuse because it is the longest side.
- 3 Double it: the chord is 24.
why: The perpendicular **bisects** the chord, so the triangle gives only half of it.

Where it goes wrong

- Answering 12 by forgetting to double.
- Adding instead of subtracting inside the square root.

Takeaway Perpendicular from centre to chord: it bisects the chord, so use Pythagoras and then double.

Question 5 FUN-03 · Functions and transformations · spec MM8.2

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $f(g(2)) - g(f(2))$.

Answer: C

$$g(2) = 1, f(1) = 3, f(2) = 5, g(5) = 22. \text{ So } 3 - 22 = -19.$$

Full method - how and why

Composite functions evaluated at a number. Work strictly from the inside out, and keep the two compositions separate.

1 $g(2) = 2^2 - 3 = 1$, then $f(1) = 2(1) + 1 = 3$. So $f(g(2)) = 3$.

why: In $f(g(2))$ the inner function acts first: feed 2 into g , then its output into f .

2 $f(2) = 2(2) + 1 = 5$, then $g(5) = 25 - 3 = 22$. So $g(f(2)) = 22$.

why: The other order gives a completely different value - composition is not commutative.

3 Subtract in the order asked: $3 - 22 = -19$.

why: The question is $f(g(2)) - g(f(2))$, so the small one comes first.

Where it goes wrong

- Computing $g(f(2)) - f(g(2))$ and answering 19.
- Applying the functions in the wrong order inside one composition.

Takeaway $f(g(x))$ means g first; check the order twice, because the wrong order is always an offered option.

Question 6 NUM-07 · Number, units and estimation · spec M1.2

A car travels at 90 km/h. What is this speed in metres per second?

Answer: D

$$90\,000 \text{ m in } 3600 \text{ s: } 90000/3600 = 25 \text{ m/s.}$$

Full method - how and why

A compound-unit conversion: convert distance and time separately.

1 $90 \text{ km} = 90 \times 1000 = 90\,000 \text{ m}$.

why: Convert the distance unit first.

2 $1 \text{ hour} = 60 \times 60 = 3600 \text{ s}$.

why: Convert the time unit.

3 Divide: $\frac{90\,000}{3600} = 25 \text{ m/s}$.

why: Equivalently, divide km/h by 3.6 - worth memorising.

Where it goes wrong

- Multiplying by 3.6 instead of dividing, giving 324 - an offered option.
- Converting only one of the two units.

Takeaway $\text{km/h} \div 3.6 = \text{m/s}$; convert numerator and denominator units separately.

Question 7 LOG-04 · Exponentials and logarithms · spec MM5.2

Given that $\log_a x = 3$ and $\log_a y = 5$, find $\log_a \left(\frac{x^2 \sqrt{y}}{a} \right)$.

Answer: B

$$2\log_a x + \frac{1}{2}\log_a y - \log_a a = 6 + \frac{5}{2} - 1 = \frac{15}{2}.$$

Full method - how and why

Break the expression apart with the log laws, then substitute the given values.

1 Split the quotient and product: $\log_a \frac{x^2 \sqrt{y}}{a} = \log_a x^2 + \log_a \sqrt{y} - \log_a a$.

why: Products become sums, quotients become differences.

2 Bring the powers down: $2\log_a x + \frac{1}{2}\log_a y - \log_a a$.

why: $\sqrt{y} = y^{1/2}$, so its log is $\frac{1}{2}\log_a y$.

3 Substitute $\log_a x = 3$, $\log_a y = 5$, $\log_a a = 1$: $6 + \frac{5}{2} - 1$.

why: $\log_a a = 1$ always - a fact worth having instantly available.

4 Add: $\frac{12+5-2}{2} = \frac{15}{2}$.

why: Put everything over 2.

Where it goes wrong

- Forgetting $\log_a a = 1$ and answering $\frac{17}{2}$.
- Using $\log_a \sqrt{y} = \sqrt{\log_a y}$.

Takeaway Expand fully with the three laws before substituting; $\log_a a = 1$ and $\log_a 1 = 0$.

Question 8 IND-21 · Indices and surds · spec MM1.1

If $x = 2^p$, $y = 2^q$ and $\frac{x^3}{y^2} = 2^5$ with $xy = 2^{10}$, find p .

Answer: F

$$3p - 2q = 5 \text{ and } p + q = 10. \text{ From the second, } q = 10 - p; \text{ then } 3p - 2(10 - p) = 5, \text{ so } 5p = 25 \text{ and } p = 5.$$

Full method - how and why

Two equations in disguise. Writing both conditions in base 2 turns them into simultaneous linear equations in p and q .

1 $\frac{x^3}{y^2} = \frac{2^{3p}}{2^{2q}} = 2^{3p-2q}$, so $3p - 2q = 5$.

why: Powers of powers multiply indices, division subtracts them; equating indices is valid because base 2 is one-to-one.

2 $xy = 2^p \cdot 2^q = 2^{p+q}$, so $p + q = 10$.

why: Multiplication adds indices.

3 Solve: from the second, $q = 10 - p$; substituting, $3p - 2(10 - p) = 5 \Rightarrow 5p = 25 \Rightarrow p = 5$.

why: Substitution is quickest with only two unknowns; note the bracket when substituting a negative.

Where it goes wrong

- Dropping the bracket to get $3p - 20 - 2p = 5$ and hence $p = 25$.
- Equating $x^3/y^2 = 2^5$ to $3p/2q = 5$ - indices subtract, they do not divide.

Takeaway Converting everything to one base turns exponential conditions into linear equations in the exponents.

Question 9 TRG-09 · Trigonometric functions and equations · spec MM4.5

Simplify $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$ for values of θ where the expression is defined.

Answer: B

$\cos \theta \tan \theta = \sin \theta$, and $1 - \cos^2 \theta = \sin^2 \theta$. So the expression is $\sin \theta$.

Full method - how and why

Simplification using the two permitted identities. Work on the denominator and numerator separately.

- 1 Denominator: $\cos \theta \tan \theta = \cos \theta \times \frac{\sin \theta}{\cos \theta} = \sin \theta$.
why: $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and the $\cos \theta$ cancels (legitimate where the expression is defined).
- 2 Numerator: $1 - \cos^2 \theta = \sin^2 \theta$.
why: Rearranged Pythagorean identity.
- 3 Divide: $\frac{\sin^2 \theta}{\sin \theta} = \sin \theta$.
why: One power of $\sin \theta$ cancels.

Where it goes wrong

- Cancelling to $\sin^2 \theta$ by forgetting to simplify the denominator.
- Writing $1 - \cos^2 \theta = \cos^2 \theta - 1$ (sign flip), giving $-\sin \theta$.

Takeaway Convert $\tan$ into $\frac{\sin}{\cos}$ and use $1 - \cos^2 = \sin^2$; then cancel.

Question 10 GRA-03 · Graphs of functions · spec MM8.1

Which of the following could be the equation of a cubic curve that touches the x -axis at $x = 2$ and crosses it at $x = -1$, and has a negative y -intercept?

Answer: A

Touching at 2 needs the factor $(x - 2)^2$ and crossing at -1 the factor $(x + 1)$. $y = (x - 2)^2(x + 1)$ has y -intercept $4 > 0$, so the sign is reversed: $y = -(x - 2)^2(x + 1)$ with intercept -4 .

Full method - how and why

Build the cubic from its root behaviour, then use the y -intercept to fix the sign.

- 1 'Touches at $x = 2$ ' means a repeated factor $(x - 2)^2$; 'crosses at $x = -1$ ' means a simple factor $(x + 1)$.
why: Even-power factors touch the axis; odd-power factors cross it.
- 2 So the cubic is $y = a(x - 2)^2(x + 1)$ for some constant $a \neq 0$.
why: Degree $2 + 1 = 3$ ✓.
- 3 Compute the y -intercept: at $x = 0$, $y = a(4)(1) = 4a$.
why: Substitute $x = 0$ into the factorised form.
- 4 A negative intercept needs $a < 0$, so $y = -(x - 2)^2(x + 1)$ works (intercept -4).
why: The option with $a = +1$ has intercept $+4$ and is the trap.

Where it goes wrong

- Choosing $(x - 2)^2(x + 1)$ and ignoring the sign of the intercept.
- Swapping which root is repeated, e.g. $(x - 2)(x + 1)^2$.

Takeaway Repeated factor $\Rightarrow$ touch; simple factor $\Rightarrow$ cross; the y -intercept fixes the leading sign.

Question 11 QUA-11 · Quadratic functions · spec MM1.3

The curve $y = x^2 - 2ax + a^2 + a - 6$ has its vertex on the x -axis. Find the value of a .

Answer: B

Complete the square: $y = (x - a)^2 + a - 6$. The vertex is $(a, a - 6)$, which lies on the x -axis when $a = 6$.

Full method - how and why

'Vertex on the x -axis' means the minimum value is zero, i.e. the quadratic has a repeated root. Completing the square is quicker than the discriminant here.

- 1 Complete the square in x , treating a as a constant: $x^2 - 2ax + a^2 = (x - a)^2$, so $y = (x - a)^2 + (a - 6)$.

why: The first three terms are already a perfect square - that is why the question is built with a^2 in it.

- 2 Read off the vertex: $(a, a - 6)$.

why: In $(x - h)^2 + k$ the vertex is (h, k) .

- 3 On the x -axis means the y -coordinate is zero: $a - 6 = 0$, so $a = 6$.

why: 'On the x -axis' is a statement about y , not x - the commonest misread in this question.

Where it goes wrong

- Setting the x -coordinate to zero and answering $a = 0$.
- Using the discriminant but mis-expanding $4a^2 - 4(a^2 + a - 6)$, which should collapse to $-4(a - 6)$.

Takeaway Vertex on the x -axis $\Leftrightarrow$ repeated root $\Leftrightarrow$ discriminant zero - three descriptions of the same situation.

Question 12 GEO-14 · Angles, polygons, congruence and vectors · spec M5.5

In triangle ABC , D lies on AB and E lies on AC with DE parallel to BC . $AD = 4$, $DB = 6$ and $DE = 5$. Find BC .

Answer: A

Triangles ADE and ABC are similar with ratio $AD : AB = 4 : 10$. $BC = 5 \times \frac{10}{4} = 12.5$.

Full method - how and why

A line parallel to one side creates a similar triangle. Use the **full** side, not just the part.

- 1 $DE \parallel BC$ makes triangles ADE and ABC similar (equal corresponding angles).

why: Parallel lines give equal corresponding angles, which is the AAA similarity condition.

- 2 Compute the ratio: $AB = AD + DB = 4 + 6 = 10$, so $AD : AB = 4 : 10 = 2 : 5$.

why: The ratio must compare corresponding **whole** sides - AD to AB , not AD to DB .

- 3 Scale up DE : $BC = 5 \times \frac{10}{4} = 12.5$.

why: Multiply by the reciprocal of the ratio to go from small to large.

Where it goes wrong

- Using the ratio $4 : 6$ and answering 7.5 - an offered option.
- Dividing instead of multiplying, giving 2.

Takeaway In similar triangles compare corresponding **full** sides: $\frac{AD}{AB}$, not $\frac{AD}{DB}$.

Question 13 SIM-03 · Simultaneous equations and inequalities · spec MM1.5

Find the set of values of x satisfying both $3x - 7 < 2$ and $x^2 > 4$.

Answer: A

$$3x < 9 \Rightarrow x < 3. \quad x^2 > 4 \Rightarrow x < -2 \text{ or } x > 2. \text{ Combining: } x < -2 \text{ or } 2 < x < 3.$$

Full method - how and why

Two inequalities joined by 'and'. Solve separately, then take the overlap - the second one produces two intervals, which is where the care is needed.

- 1 Solve the linear one: $3x - 7 < 2 \Rightarrow 3x < 9 \Rightarrow x < 3$.
why: Dividing by a positive number leaves the inequality direction unchanged.
- 2 Solve the quadratic one: $x^2 > 4 \Rightarrow |x| > 2 \Rightarrow x < -2 \text{ or } x > 2$.
why: $x^2 > 4$ is satisfied by large positive **and** large negative x - taking only the positive branch is the classic error.
- 3 Intersect the sets on a number line: from $x < 3$, keep the parts of $x < -2$ (all of it) and of $x > 2$ (only up to 3).
why: 'Both' means intersection; drawing both solution sets on a single line prevents mistakes.
- 4 Answer: $x < -2 \text{ or } 2 < x < 3$.
why: The answer is itself two intervals - a perfectly legitimate solution set.

Where it goes wrong

- Dropping the branch $x < -2$ and answering $2 < x < 3$.
- Solving $x^2 > 4$ as $x > \pm 2$, which is not a meaningful statement.

Takeaway $x^2 > k$ gives two intervals; intersect solution sets on a number line rather than in your head.

Question 14 SEQ-07 · Sequences and series · spec MM2.1

A sequence satisfies $x_{n+1} = \frac{1}{1-x_n}$ with $x_1 = 2$. Find x_{100} .

Answer: B

$$x_2 = \frac{1}{1-2} = -1, \quad x_3 = \frac{1}{1-(-1)} = \frac{1}{2}, \quad x_4 = \frac{1}{1-\frac{1}{2}} = 2 = x_1: \text{ the sequence has period 3. Since } 100 = 3 \times 33 + 1, \\ x_{100} = x_1 = 2.$$

Full method - how and why

A recurrence with a huge index. Compute a few terms, spot the cycle, then use arithmetic on the index.

- 1 Iterate: $x_2 = \frac{1}{1-2} = -1$, $x_3 = \frac{1}{1-(-1)} = \frac{1}{2}$, $x_4 = \frac{1}{1-\frac{1}{2}} = 2$.
why: When an index like 100 appears, always compute the first few terms - the sequence is almost certainly periodic.
- 2 $x_4 = x_1$, so the sequence repeats with period 3: $2, -1, \frac{1}{2}, 2, -1, \frac{1}{2}, \dots$
why: Once a value repeats, the whole sequence from that point repeats identically.
- 3 Reduce the index: $100 = 3 \times 33 + 1$, so x_{100} is in the same position of the cycle as x_1 .
why: Terms with indices differing by a multiple of the period are equal; compare remainders on division by 3.
- 4 Therefore $x_{100} = x_1 = 2$.
why: Check the convention: the indices 1, 4, 7, ... all leave remainder 1 on division by 3, and 100 does too.

Where it goes wrong

- Finding that 100 leaves remainder 1 but then reading off x_0 or x_3 - line the remainders up with the actual starting index.
- Giving up after two terms and guessing -1 or $\frac{1}{2}$.

Takeaway Large-index recurrence questions are periodicity questions: find the cycle length, then work with the index modulo it.

Question 15 CIR-16 · Circles and circle theorems · spec MM3.2

The point $P(x, y)$ moves so that its distance from $(3, 0)$ is twice its distance from $(0, 0)$. The locus of P is a circle. Find its radius.

Answer: C

$$(x - 3)^2 + y^2 = 4(x^2 + y^2) \Rightarrow 3x^2 + 3y^2 + 6x - 9 = 0 \Rightarrow x^2 + 2x + y^2 = 3 \Rightarrow (x + 1)^2 + y^2 = 4. \text{ Radius } 2.$$

Full method - how and why

A locus defined by a distance ratio. Write the condition algebraically, square to remove roots, and complete the square.

1 Write the condition: $\sqrt{(x - 3)^2 + y^2} = 2\sqrt{x^2 + y^2}$.

why: 'Distance from A is twice the distance from B' - write both distances with the formula.

2 Square both sides: $(x - 3)^2 + y^2 = 4(x^2 + y^2)$.

why: Squaring is safe because both sides are non-negative; and it removes both roots at once. Note the right-hand side gains a factor 4, not 2.

3 Expand and collect: $x^2 - 6x + 9 + y^2 = 4x^2 + 4y^2 \Rightarrow 3x^2 + 3y^2 + 6x - 9 = 0$.

why: Bring everything to one side.

4 Divide by 3 and complete the square: $x^2 + 2x + y^2 = 3 \Rightarrow (x + 1)^2 + y^2 = 4$, so $r = 2$.

why: Dividing by the common factor first makes the coefficients of x^2 and y^2 equal to 1 - necessary before completing the square.

Where it goes wrong

- Squaring the right side as $2(x^2 + y^2)$ instead of $4(x^2 + y^2)$.
- Forgetting to divide by 3 before completing the square, which corrupts the radius.

Takeaway Distance-ratio loci (Apollonius circles): square the condition, divide so x^2 and y^2 have coefficient 1, then complete the square.

Question 16 ALG-03 · GCSE algebra · spec M4.6

Simplify $\frac{x^2 - 9}{x^2 + x - 12} \times \frac{x^2 - 16}{x^2 - 3x}$.

Answer: A

Factorise everything: $\frac{(x - 3)(x + 3)}{(x + 4)(x - 3)} \times \frac{(x - 4)(x + 4)}{x(x - 3)}$. The $(x + 4)$ factors cancel and one $(x - 3)$ cancels, leaving $\frac{(x + 3)(x - 4)}{x(x - 3)}$.

Full method - how and why

Multiplying algebraic fractions: factorise everything first, then cancel common factors across the product.

1 Factorise each part: $x^2 - 9 = (x - 3)(x + 3)$; $x^2 + x - 12 = (x + 4)(x - 3)$; $x^2 - 16 = (x - 4)(x + 4)$; $x^2 - 3x = x(x - 3)$.

why: Nothing can be cancelled until everything is in factor form.

2 Write the product: $\frac{(x - 3)(x + 3)}{(x + 4)(x - 3)} \times \frac{(x - 4)(x + 4)}{x(x - 3)}$.

why: In a product of fractions any top factor can cancel with any bottom factor.

3 Cancel: the $(x + 4)$ pair, and one $(x - 3)$ from the first numerator with the first denominator.

why: There are two $(x - 3)$ s on the bottom but only one on top, so one survives below.

4 What remains: $\frac{(x + 3)(x - 4)}{x(x - 3)}$.

why: Count the leftovers carefully - this is where marks are lost.

Where it goes wrong

- Cancelling both $(x - 3)$ factors and losing one from the denominator.
- Cancelling terms rather than factors, e.g. striking out the x^2 s.

Takeaway Factorise fully, then cancel factor-by-factor and count how many copies of each remain.

Question 17 DIF-08 · Differentiation · spec MM6.2

Given $f(x) = x^{\frac{3}{2}} - 6x^{\frac{1}{2}}$ for $x > 0$, find the value of x at which f has its minimum.

Answer: A

$f'(x) = \frac{3}{2}x^{1/2} - 3x^{-1/2} = \frac{3}{2\sqrt{x}}(x - 2) = 0$ at $x = 2$; f' changes from negative to positive there, so it is a minimum.

Full method - how and why

Differentiating fractional powers. Factorising the derivative makes the sign change easy to see.

- 1 Differentiate: $f'(x) = \frac{3}{2}x^{1/2} - 3 \times \frac{1}{2}x^{-1/2} = \frac{3}{2}x^{1/2} - 3x^{-1/2}$.
why: $\frac{d}{dx}x^{3/2} = \frac{3}{2}x^{1/2}$ and $\frac{d}{dx}(6x^{1/2}) = 3x^{-1/2}$.
- 2 Factor out $\frac{3}{2\sqrt{x}}$: $f'(x) = \frac{3}{2\sqrt{x}}(x - 2)$.
why: $\frac{3}{2}x^{1/2} = \frac{3}{2\sqrt{x}} \cdot x$ and $3x^{2\sqrt{x}} = \frac{3}{2\sqrt{x}} \cdot 2$. Factorising isolates the sign-changing part.
- 3 Set $f' = 0$: since $\frac{3}{2\sqrt{x}} > 0$ for $x > 0$, the only root is $x = 2$.
why: A positive factor can never make the product zero.
- 4 Check it is a minimum: f' is negative for $x < 2$ and positive for $x > 2$, so the curve falls then rises.
why: The sign of f' is the sign of $(x - 2)$ - immediate from the factorised form.

Where it goes wrong

- Making index errors: $\frac{d}{dx}x^{1/2} = \frac{1}{2}x^{-1/2}$, not $\frac{1}{2}x^{1/2}$.
- Squaring both sides carelessly when solving $\frac{3}{2}\sqrt{x} = \frac{3}{\sqrt{x}}$ and getting $x = 4$.

Takeaway Factor a common power out of the derivative: the remaining bracket controls the sign.

Question 18 QUA-20 · Quadratic functions · spec MM1.3

For how many integer values of k does $x^2 + kx + 12 = 0$ have integer roots?

Answer: F

Integer roots r, s with $rs = 12$: pairs (1, 12), (2, 6), (3, 4) and their negatives, giving $k = -(r + s) = -13, -8, -7, 13, 8, 7$: six values.

Full method - how and why

Integer roots means an integer factorisation of the constant term. Count systematically and do not forget negative pairs.

- 1 If the roots are integers r and s then $rs = 12$ and $r + s = -k$.
why: For a monic quadratic, product of roots = c and sum = $-b$.
- 2 List the positive factor pairs of 12: (1, 12), (2, 6), (3, 4).
why: Work through divisors in order so none is missed.
- 3 Each pair also has a negative version $(-1, -12), (-2, -6), (-3, -4)$, since two negatives also multiply to +12.
why: The product is positive, so the roots share a sign - both positive or both negative.
- 4 Each of the six pairs gives a different $k = -(r + s)$: $-13, -8, -7, 13, 8, 7$. So six values.
why: Check they are distinct before counting.

Where it goes wrong

- Listing only the positive pairs and answering 3.
- Counting (1, 12) and (12, 1) as different, which double-counts.

Takeaway Integer-root problems become divisor-counting problems; the sign of the product tells you whether the roots share a sign.

Question 19 INT-14 · Integration · spec MM7.5

The trapezium rule is used to estimate $\int_a^b f(x) dx$. For which of the following functions is the estimate guaranteed to be an underestimate for every choice of $a < b$?

Answer: A

The trapezium rule underestimates when the curve is concave (bends downwards) so that chords lie below the curve: $y = -x^2$ everywhere. x^3 changes concavity at 0.

Full method - how and why

A conceptual question about the trapezium rule: it is the curve's concavity, not its sign or steepness, that decides.

- 1 Recall the rule: the trapezium rule replaces the curve by straight chords.

why: Each strip's true area is compared with a trapezium whose top is the chord.

- 2 A chord lies **below** a concave (downward-bending) curve, so the estimate is too small: an underestimate.

why: Concave means $f'' < 0$ throughout - the curve bulges above its chords.

- 3 Check the options: $-x^2$ has $f'' = -2 < 0$ everywhere ✓. x^2 and x^4 are convex (overestimate); $2x + 1$ is straight (exact); x^3 changes concavity at 0, so it depends on the interval.

why: 'Guaranteed for every choice of $a < b$ ' rules out x^3 , whose behaviour varies.

- 4 So $f(x) = -x^2$.

why: The only universally concave option.

Where it goes wrong

- Choosing x^3 without noticing that it is concave for $x < 0$ but convex for $x > 0$.
- Thinking a negative function must give an underestimate - sign is irrelevant; shape is everything.

Takeaway Trapezium rule: concave ($f'' < 0$) underestimates, convex ($f'' > 0$) overestimates, linear is exact.

Question 20 BIN-15 · Binomial expansion · spec MM2.4

The coefficient of x^2 in $(1 + x)^n + (1 + x)^{n+1}$ is 36. Find n .

Answer: C

$$\binom{n}{2} + \binom{n+1}{2} = \frac{n(n-1) + (n+1)n}{2} = n^2 = 36, \text{ so } n = 6.$$

Full method - how and why

Two expansions added. Write both coefficients in factorial form and the sum simplifies remarkably.

- 1 The x^2 coefficients are $\binom{n}{2}$ and $\binom{n+1}{2}$.

why: One from each expansion; the total coefficient of x^2 is their sum.

- 2 Write them out: $\frac{n(n-1)}{2} + \frac{(n+1)n}{2} = \frac{n[(n-1) + (n+1)]}{2}$.

why: Factor out $\frac{n}{2}$ - the brackets then collapse.

- 3 Simplify: $\frac{n \times 2n}{2} = n^2$.

why: A surprisingly clean result; recognising it makes the final step trivial.

- 4 Solve $n^2 = 36$, so $n = 6$ (positive integer).

why: n counts an index, so the negative root is rejected.

Where it goes wrong

- Using Pascal's rule to write the sum as $\binom{n+1}{3}$ - that rule adds **adjacent** entries in the same row, not the same entry in different rows.
- Taking $n = \pm 6$ and offering -6 .

Takeaway $\binom{n}{2} + \binom{n+1}{2} = n^2$: write coefficients in factorial form and look for common factors before solving.